

tolerance levels and leaving the MGS dead in space.

System Configuration

Getting to the meaty part for us *Digit-ians*, we want to know what kind of computers are used in spacecraft. We'll take a look at what powers the two rovers currently on Mars to give you an idea.

If you're expecting a super-computer, you're going to be very disappointed. The computational power on a rover is about as good as the average budget laptop! Why? Take a glance back at *Power Generation* above...

The CPU used in the rovers was a specially designed 32-bit Rad 6000 processor, which is shielded against radiation and is based on the PowerPC architecture—yes, like the ones the older Apple Macs used to run. The CPU is capable of a modest 20 million calculations per second, and is supported by a mere 128 MB of RAM. There's also 256 MB of Flash memory and various other smaller amounts of non-volatile memory. The system is designed specially to waste almost no energy in heat dissipation, and is completely sealed inside what NASA calls a "Warm Electronics Box" (WEB).

Unlike here on Earth, in space, computers have to be protected against radiation and extreme cold. Considering that the temperature of an average Martian night is about minus 90 degrees Celsius, the air is filled with rust-dust and the atmosphere offers no protection against solar radiation (no ozone layer or even air to help out), it's no wonder the "brains" of the rovers are sealed in the WEB.

The walls of the WEB are painted with gold to reflect radiation; the box itself is completely airtight and insulated with aerogel—a compound that's made up of 99.8 per cent air and 0.2 per cent silica dioxide, is light as a feather, extremely solid and is sometimes referred to as "solid smoke" by NASA engineers.

To give you a better idea about how well aerogel can insulate, consider this: you could take a thin sheet of aerogel, hold a blowtorch under one end and actually place your hand on the other end without feeling anything more than "warmth." It's "solid" too—a slab of aerogel as big as an adult human weighs a mere 455 gm, and can support an object that weighs 500 kg! Don't expect to see it in your local hardware store anytime soon though—it costs way too much to create. Still, if you have money to burn, hop on over to www.aerogel.com and see what you can afford.

Software And Navigation

When man does finally go to Mars, he will not be alone. Space exploration relies heavily on robots and vehicles that work as autonomous entities. Computers need to just calculate and work instead of sitting around dumb, the way the average PC does, waiting to be told

what to do. During lift-off and early flight, computers do almost everything to make sure that the rocket stays on course. During landing on Mars as well, computers have proven to be quite capable—again, we'll cite the example of the two rovers that were successfully landed on Mars. Humans on Mars will use vehicles to get to and from exploration sites there, and these vehicles cannot be as dumb as the average road car. Let's

look at what makes the Mars rovers tick to better understand what will power future landings on Mars—manned or otherwise.

Carnegie Mellon University developed the "navigation and hazard avoidance software" (NHAS) that the two rovers depend on to keep themselves out of trouble on Mars. Using this software, the rovers have achieved an average autonomous driving speed of 34 metres an hour. Before you laugh-out-loud incredulously, exclaiming

"34 metres an hour?", you should understand the complexities involved.

Mars is a crater-ridden, rock strewn, dusty and stormy planet. With volcanoes three times the size of Mt. Everest, and canyons that dwarf the Grand Canyon, and billions of rocks on its surface, Mars is even the bravest off-road driver's worst nightmare. (Damn those Martians—why couldn't they build highways!) Add to that the fact that each rover is carrying rather expensive and critical information and equipment, and you should understand why the rovers don't just race along the Martian surface, bouncing off boulders sending back messages to Earth that say "Whee, this is fun!"

Each rover is equipped with pairs of stereo cameras to give depth perception similar to what the human eyes afford, and click pictures of the paths in front of them. When the rovers are given a destination (in coordinates) by technicians on Earth, they first identify the destination, clicking pictures of the straight-line path there. Next they start identifying obstacles on the path using the NHAS developed at Carnegie Mellon to examine the high resolution stereo images they click and plot multiple paths. After all calculations, permutations and combinations are done, the rover identifies the shortest path of least resistance (lowest hazard risk path) and then moves between half and two metres. Once it's moved, it re-clicks stereo images and the whole computation process is repeated. Incidentally, the rovers are capable of identifying and measuring the distance to about 16,000 points in each stereo photograph they click!

Calculating their own movement is not left to simple odometers either. (For those who don't know, an odometer is a device found on cars and motorcycles that calculates the speed and distance traversed by a wheel; it measures this by calculating how fast the wheel is turning and how many times it has



Aerogel's almost not even there, yet it's stronger than most



A Mars rover hard at work



On the left is a before, on the right is during a dust storm—now that's a storm!